

2012 Revised Higher Physics: scan the code and answer these.



Q17

Q18

Q20

Section A · electricity

Show your working, not just the letter.

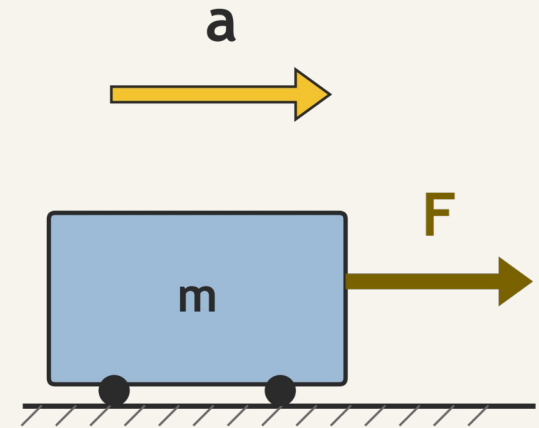


Scan for the paper
Marking scheme

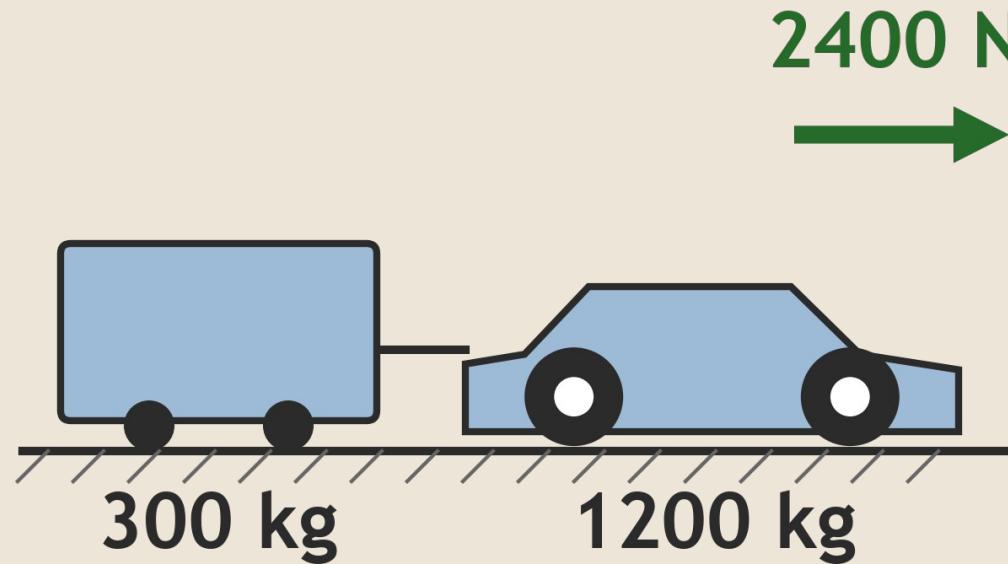
Forces

Why does anything ever speed up?

Higher Physics



A car tows a trailer. Is the pull on the trailer as big as the car's engine force?



friction: car 300 N,
trailer 150 N

PART 1

Balanced and unbalanced forces

Motion only changes when the forces stop balancing.

By the end of this part...



- 1 I can add forces as vectors to find the resultant force
- 2 I can use Newton's first and second laws to explain motion
- 3 I can use $F = ma$ with the resultant of several forces

Recap from N5: Forces

Balanced

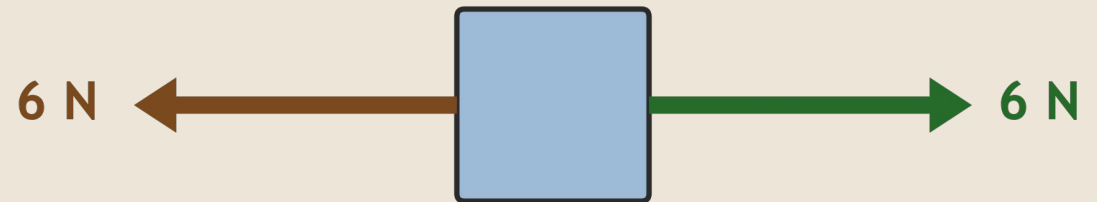
at rest or constant velocity

Unbalanced

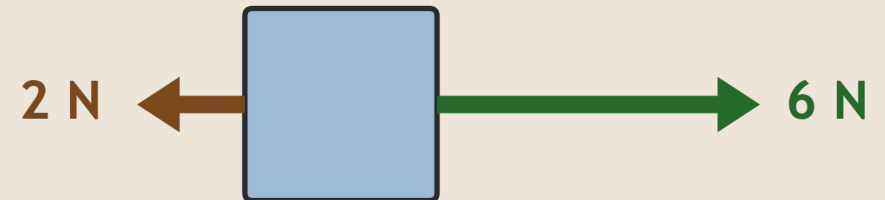
accelerates

Free body diagram

only the forces ON the object



balanced: resultant = 0



unbalanced: resultant = 4 N →

Recap: $F = ma$

$$F = ma$$

F

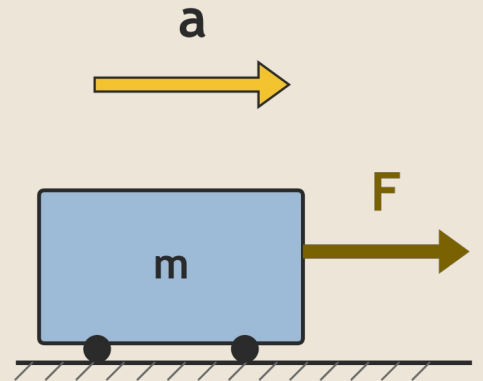
unbalanced
force (N)

m

mass (kg)

a

acceleration
(m s^{-2})



Recap: $W = mg$

$$W = mg$$

W

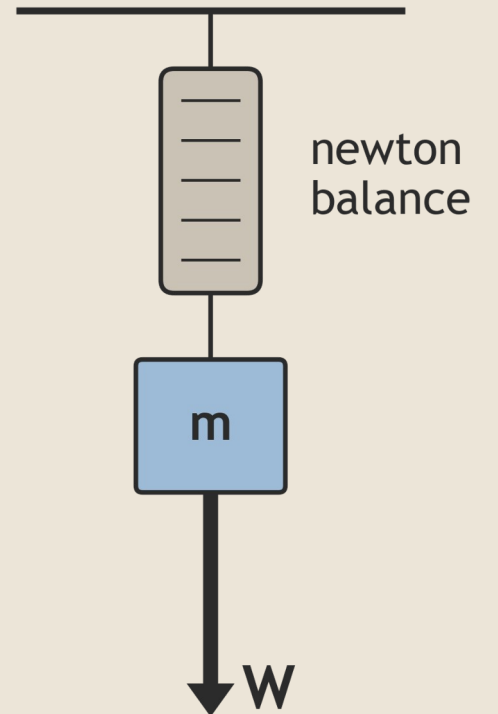
weight (N)

m

mass (kg)

g

gravitational
field strength
(N kg^{-1})



Worked Example: Forces at Right Angles

Ropes pull a 50 kg crate across smooth ice: 30 N east and 40 N north. Calculate the magnitude of acceleration.

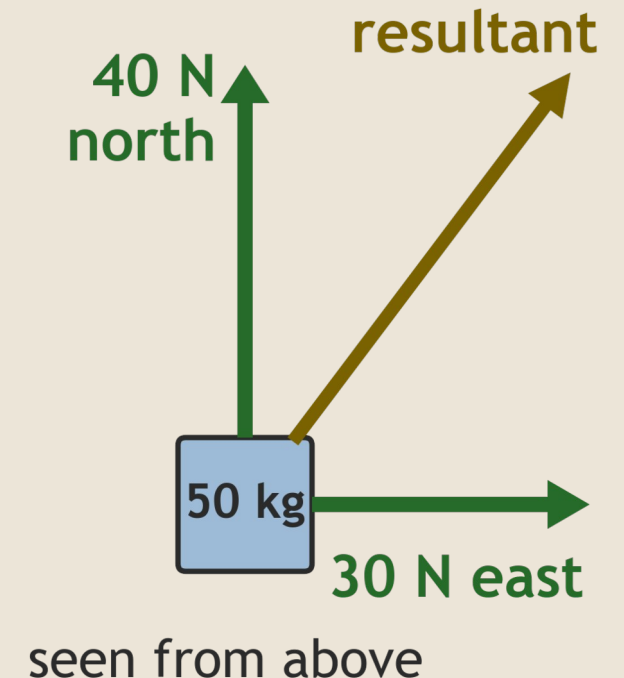
4

$$F = ma \quad (1)$$

$$F_R = \sqrt{30^2 + 40^2} = 50 \text{ N} \quad (1)$$

$$50 = 50 \times a \quad (1)$$

$$a = 1.0 \text{ m s}^{-2} \quad (1)$$



Worked Example: A Falling Ball

A 0.50 kg ball falls. Air resistance is 1.9 N. Calculate its acceleration.

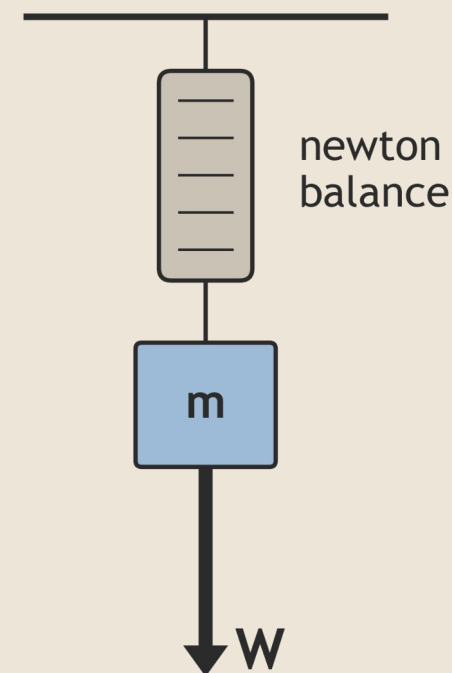
3

$$F = ma \quad (1)$$

$$(W = mg = 0.50 \times 9.8 = 4.9 \text{ N})$$

$$4.9 - 1.9 = 0.50 \times a \quad (1)$$

$$a = 6.0 \text{ m s}^{-2} \quad (1)$$



Your Turn: Forces



1

6.0 N east and 8.0 N north act on a 2.0 kg mass. Calculate its acceleration.

5.0 m s⁻² — so the resultant is 10 N

2

A 0.50 kg ball falls. Air resistance is 2.9 N. Calculate its acceleration.

4.0 m s⁻²

3

A 1200 kg car accelerates at 1.5 m s⁻². Friction is 600 N. Calculate the engine force.

2400 N — so ma + friction

2025 Paper 1 Q3. A force of 10.0 N pulls a trolley of mass 2.0 kg along a horizontal surface. The total frictional force on the trolley is 6.0 N. The acceleration is



A 2.0 m s^{-2}

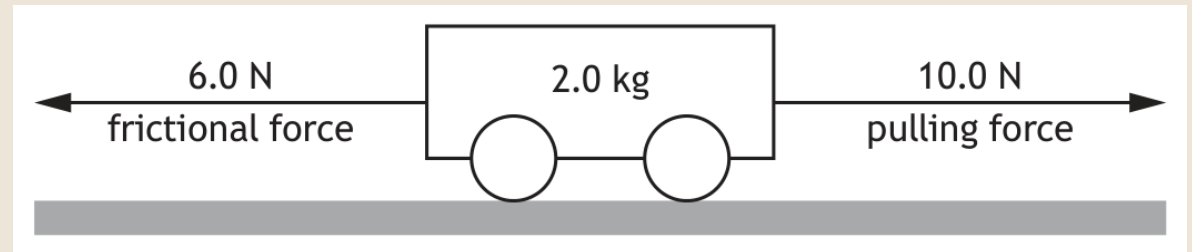
B 3.0 m s^{-2}

C 5.0 m s^{-2}

D 8.0 m s^{-2}

E 10.0 m s^{-2}

Answer A: $F = 10.0 - 6.0 = 4.0 \text{ N}$ — so $a = 4.0 \div 2.0$

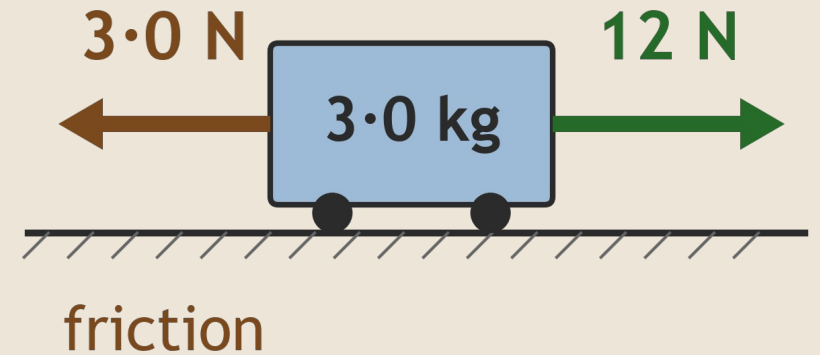


Spot the mistake. A 3.0 kg trolley is pulled by 12 N. Friction is 3.0 N. What is wrong with this answer?



$$a = F \div m$$
$$a = 12 \div 3.0$$
$$a = 4.0 \text{ m s}^{-2}$$

F is the unbalanced force – so $F = 9.0$
N and $a = 3.0 \text{ m s}^{-2}$



Plenary: Forces and $F = ma$



1

5.0 N and 12 N act at right angles. Calculate the resultant.

13 N

2

Which force goes in $F = ma$?

The resultant force

3

A car moves at constant velocity. What is the resultant force?

Zero – so balanced

Look back — by the end of this part...



- 1 I can add forces as vectors to find the resultant force
- 2 I can use Newton's first and second laws to explain motion
- 3 I can use $F = ma$ with the resultant of several forces

2015 Higher Physics: scan the code and answer these.



Q8

Q10

Q16

Section A · particles, a.c., uncertainty

Show your working, not just the letter.



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Marking scheme

PART 2

Newton's laws in action

Two blocks, one string: which mass do you use?

By the end of this part...



- 1 I can draw free body diagrams for connected objects
- 2 I can find the tension between connected objects, with friction

Connected Objects

System mass $m_1 + m_2$

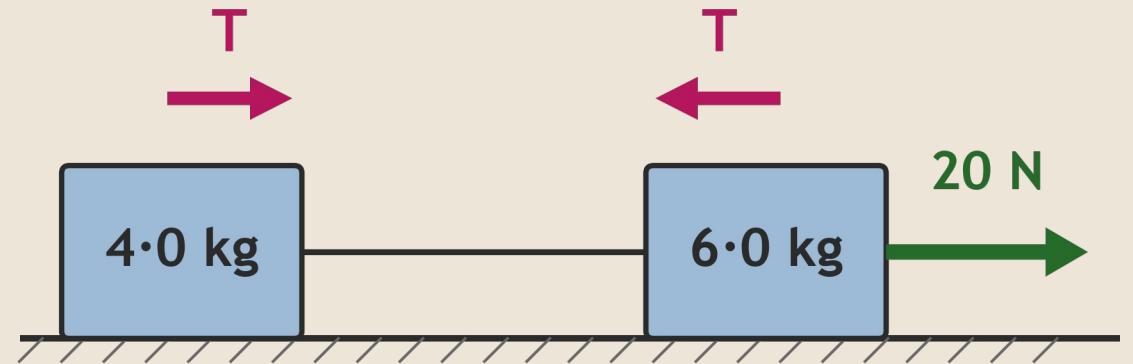
$$F = (m_1 + m_2) a$$

Tension T

the pull of the string on each block

One block

use only that block's mass



Connected Objects: The Method



1 First find a , using the whole system:

$$F = (m_1 + m_2) a$$

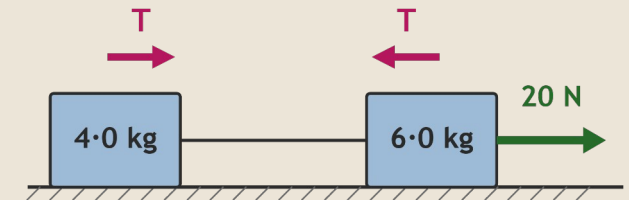
2 Then find T , using one block:

$$T - \text{friction} = ma$$

F = unbalanced force = pull – friction

$m_1 + m_2$ = total mass, m = one block's mass

T = tension in the string



Worked Example: Two Trolleys

10 N pulls the 2.0 kg trolley. No friction. Calculate the tension in the string.

4

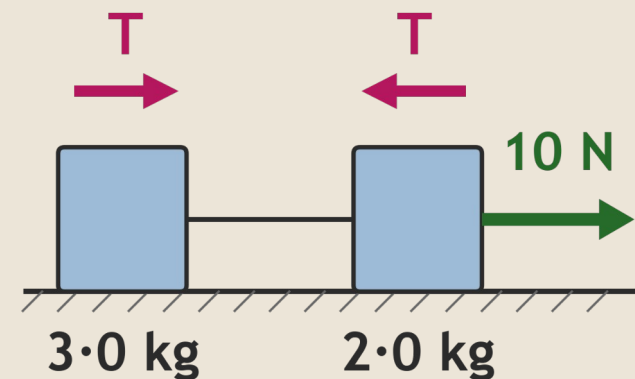
$$F = ma \quad (1)$$

$$10 = (3.0 + 2.0) \times a \quad (1)$$

$$a = 2.0 \text{ m s}^{-2}$$

$$T = 3.0 \times 2.0 \quad (1)$$

$$T = 6.0 \text{ N} \quad (1)$$



Worked Example: Tension with Friction

A car tows a trailer with the forces shown. Calculate the tension in the tow bar.

4

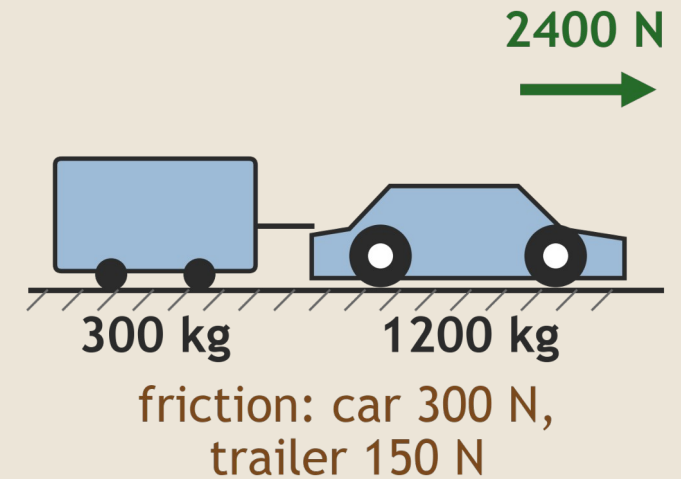
$$F = ma \quad (1)$$

$$2400 - 450 = 1500 \times a \quad (1)$$

$$(a = 1.3 \text{ m s}^{-2})$$

$$T - 150 = 300 \times 1.3 \quad (1)$$

$$T = 540 \text{ N} \quad (1)$$



Your Turn: Connected Objects



1 A 1000 kg tractor pulls a 500 kg trailer with a 900 N force. No friction. Calculate the tension.

300 N

2 The same, but 100 N friction acts on the trailer only.

370 N – so add the friction

3 10 N pulls a 3.0 kg block, which pulls a 2.0 kg block. No friction. Calculate T.

4.0 N – so the 2.0 kg block's mass

2015 Section 1 Q5. A 900 kg car pulls a 400 kg caravan with an acceleration of 2.0 m s^{-2} . Friction on the caravan is negligible. The tension in the coupling is



A 400 N

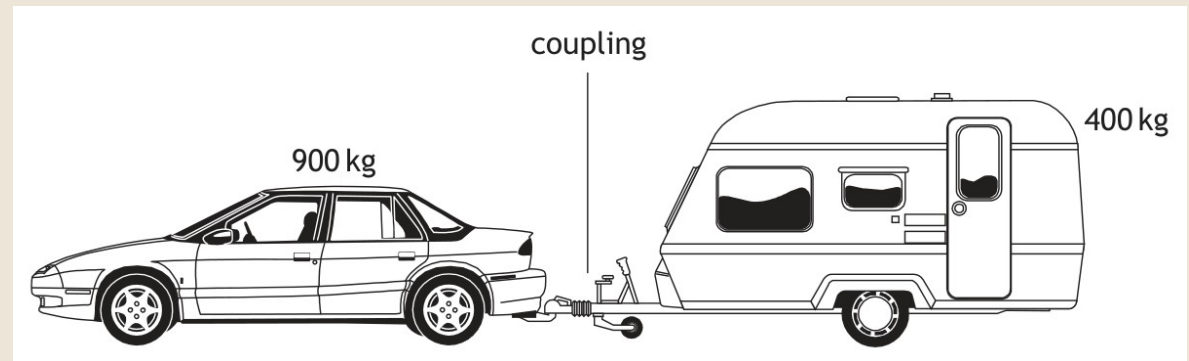
B 500 N

C 800 N

D 1800 N

E 2600 N

Answer C: $T = 400 \times 2.0$ — so the caravan's mass only



Spot the mistake. A 1200 kg car tows a 600 kg trailer at 0.50 m s^{-2} . No friction on the trailer. What is wrong with this answer?



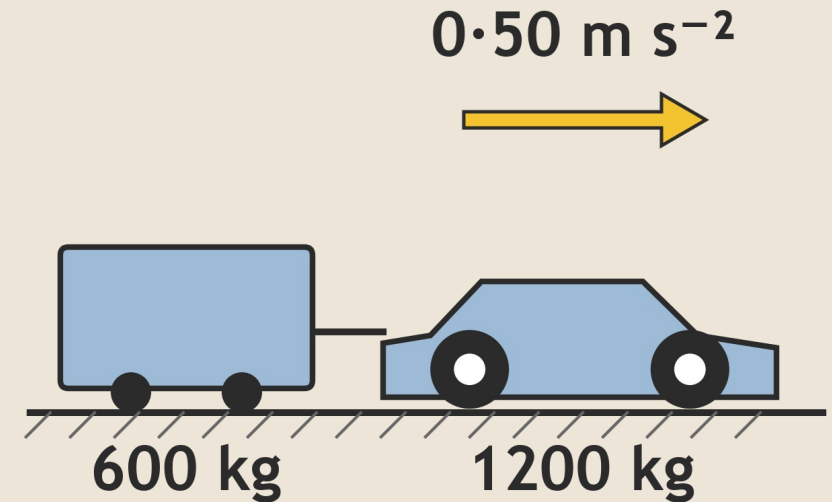
$$T = ma$$

$$T = (1200 + 600) \times 0.50$$

$$T = 900 \text{ N}$$

The tow bar pulls only the trailer – so

$$T = 600 \times 0.50 = 300 \text{ N}$$



Plenary: Newton's Laws in Action



1

2.0 kg and 3.0 kg blocks are pulled by 10 N. No friction. Calculate a .

2.0 m s^{-2}

2

For the tension in a string, which mass do you use?

Only the block the string pulls

3

Friction acts on a towed trailer. Does T go up or down?

Up — so $T = ma + \text{friction}$

Look back — by the end of this part...



- 1 I can draw free body diagrams for connected objects
- 2 I can find the tension between connected objects, with friction

2014 Higher Physics: scan the code and answer these.



Q2

Q3

Q9

Section A · motion, circuits

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 3

Lifts and apparent weight

Same pupil, same mass. The scales show three different readings.

By the end of this part...



- 1 I can draw the forces on a person standing on scales in a lift
- 2 I can use $F = ma$ to find a scales reading or a lift's acceleration
- 3 I can describe how to measure a lift's acceleration with newton scales

Why do you feel heavier as a lift starts to go up?



A lift



A drop tower ride

What the Scales Read

N = normal force

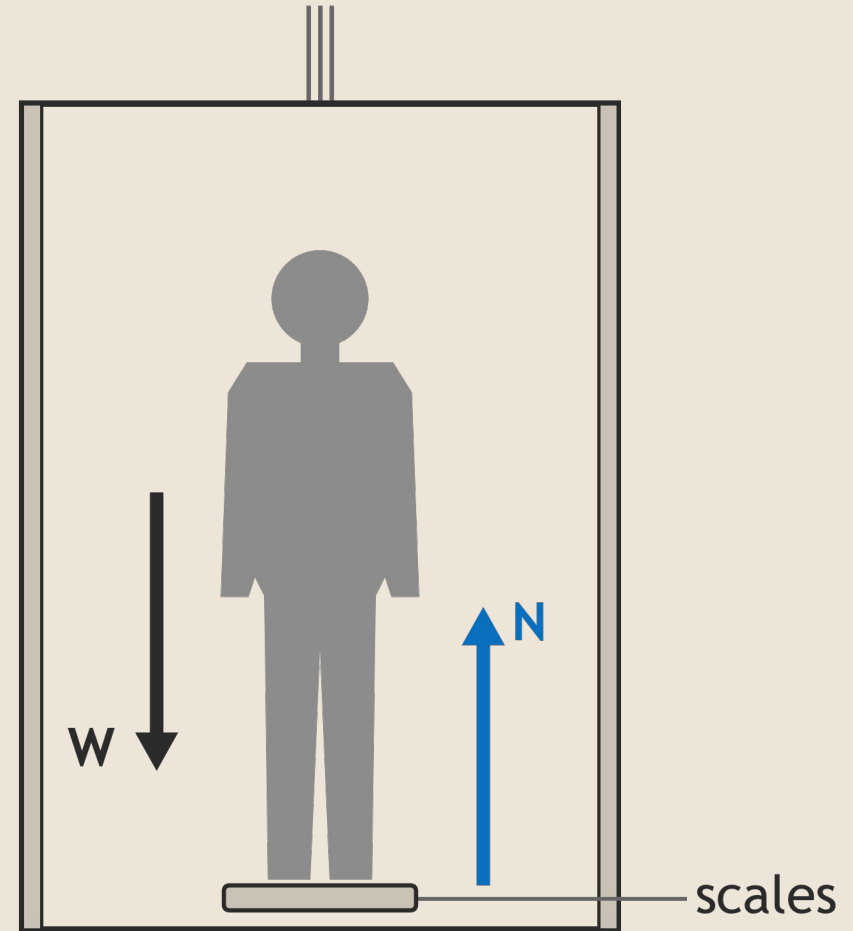
the scales push up on you

W = weight

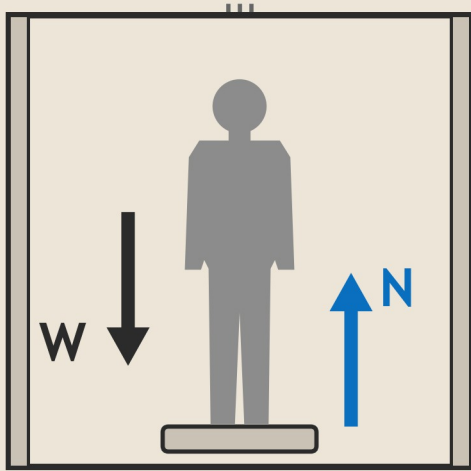
$W = mg$, down, never changes

Scales

read N (a newton balance)



A Lift Going Up



$$v = 0$$

$$a = 0$$

$$N = 588 \text{ N}$$

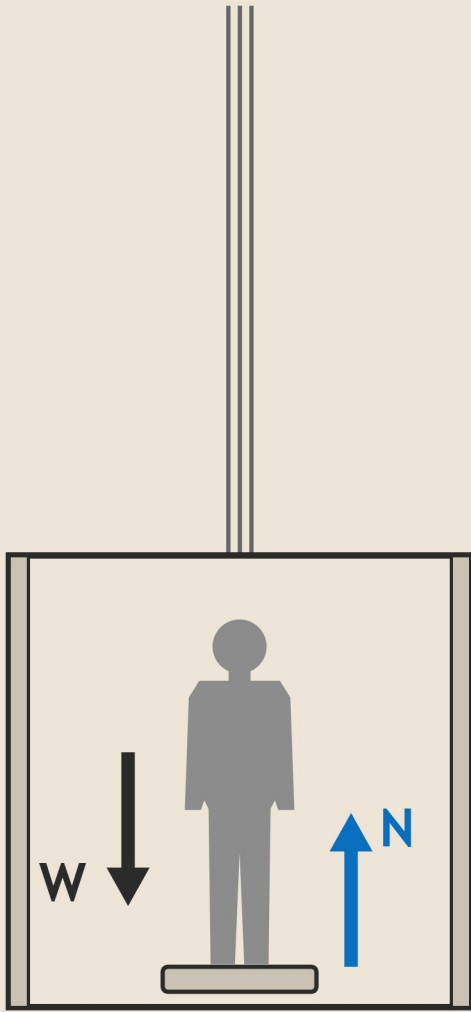
$$W = 588 \text{ N}$$

- 1 At rest: $N = W$.
- 2 Speeding up: $N > W$.
- 3 Steady speed: $N = W$.
- 4 Slowing down: $N < W$.
- 5 Stopped: $N = W$.

v = velocity

a = acceleration

A Lift Going Down



$$v = 0$$

$$a = 0$$

$$N = 588 \text{ N}$$

$$W = 588 \text{ N}$$

1 Speeding up going down: $N < W$.

2 Steady speed: $N = W$.

3 Slowing down: $N > W$.

4 Stopped: $N = W$.

v = velocity

a = acceleration

Summary: Up and Down



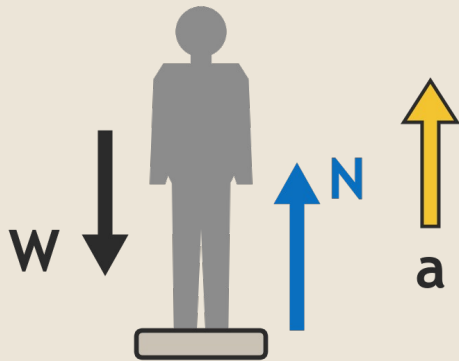
N = normal force (the scales reading) W = weight

	Going up	Going down
Acceleration	$N > W$	$N < W$
Deceleration	$N < W$	$N > W$
Constant velocity	$N = W$	$N = W$

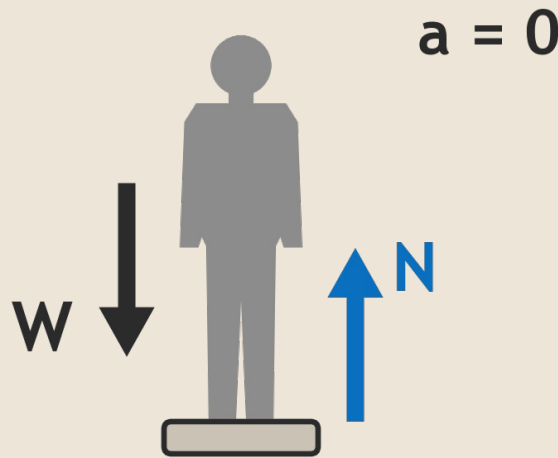
Which Way Is a?



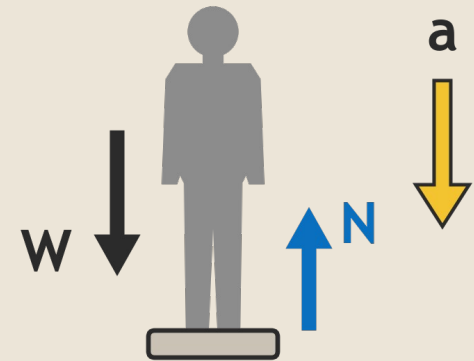
Take up as positive: $N - W = ma$ ($m = \text{mass}$)



a up: $N > W$
speeding up going
up, or slowing
going down



a = 0: $N = W$
at rest, or steady
speed



a down: $N < W$
speeding up going
down, or slowing
going up

One Ride Up, Two Graphs

Speeding up

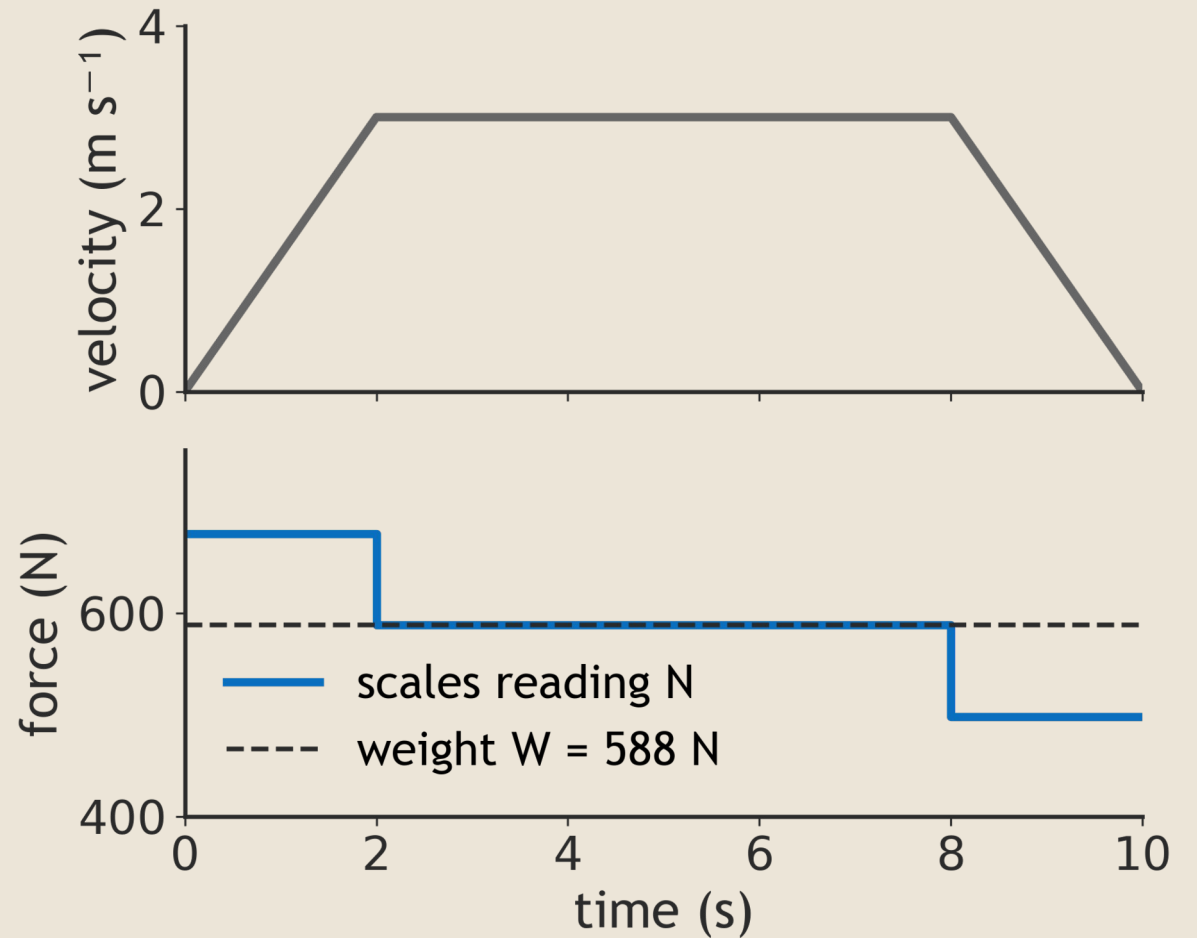
v-t line rises: N above W

Steady

v-t line flat: $N = W$

Slowing down

v-t line falls: N below W



Worked Example: Accelerating Up

A 60 kg pupil is on scales in a lift accelerating up at 1.5 m s^{-2} . Calculate the reading.

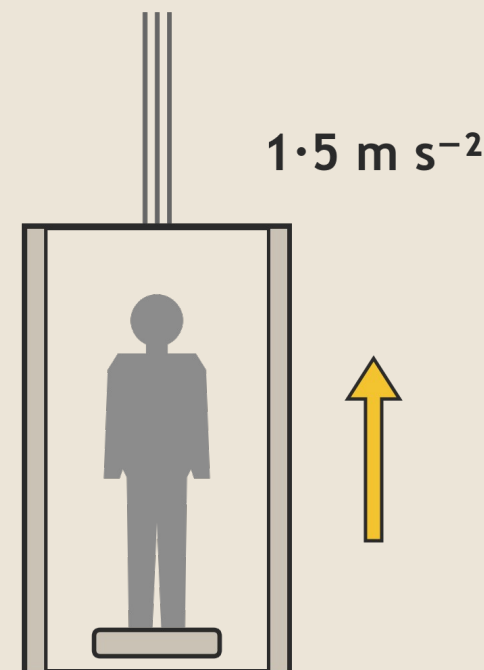
3

$$F = ma \quad (1)$$

$$(W = mg = 60 \times 9.8 = 588 \text{ N})$$

$$N - 588 = 60 \times 1.5 \quad (1)$$

$$N = 678 \text{ N} \quad (1)$$



Worked Example: Accelerating Down

A 45 kg pupil is on scales in a lift accelerating down at 2.0 m s^{-2} . Calculate the reading.

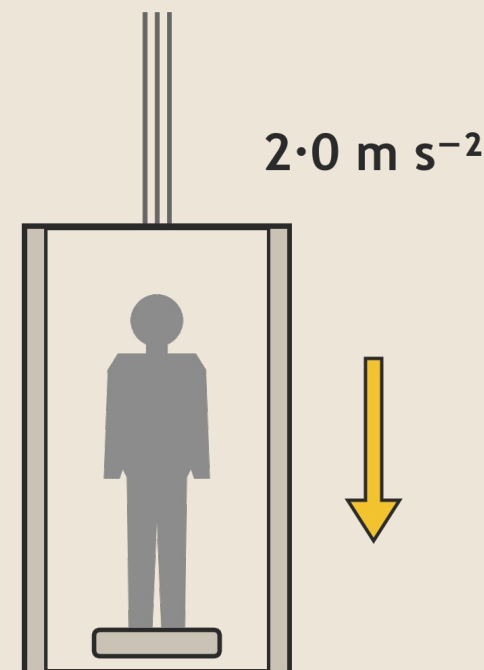
3

$$F = ma \quad (1)$$

$$(W = mg = 45 \times 9.8 = 441 \text{ N})$$

$$N - 441 = 45 \times (-2.0) \quad (1)$$

$$N = 351 \text{ N} \quad (1)$$



Worked Example: Finding a

Scales in a lift read 560 N for a 50 kg pupil. Calculate the lift's acceleration.

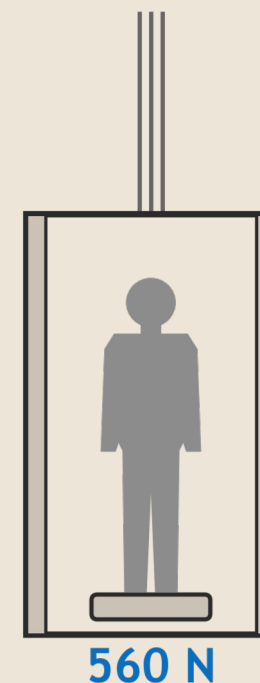
3

$$F = ma \quad (1)$$

$$(W = mg = 50 \times 9.8 = 490 \text{ N})$$

$$560 - 490 = 50 \times a \quad (1)$$

$$a = 1.4 \text{ m s}^{-2} \text{ upwards} \quad (1)$$



Your Turn: Scales in a Lift



1

A 70 kg pupil. The lift accelerates upwards at 0.80 m s^{-2} . Calculate the reading.

742 N

2

A 55 kg pupil. The lift slows down at 1.2 m s^{-2} going up. Calculate the reading.

473 N – so a is downwards, $N < W$

3

The scales read 0 N. State how the lift is accelerating.

Downwards at 9.8 m s^{-2} – so falling freely

2015 Section 1 Q4. A person stands on bathroom scales in a lift. The scales show a reading greater than the person's weight. The lift is moving



- A upwards with constant speed
- B downwards with constant speed
- C downwards with increasing speed
- D downwards with decreasing speed**
- E upwards with decreasing speed.

Answer D: slowing down going down – so a is upwards and $N > W$

2022 Paper 1 Q5. A student of mass 50.0 kg stands on a newton balance in a lift. The lift accelerates upwards at 1.2 m s^{-2} . The reading on the newton balance is:



A 60 N

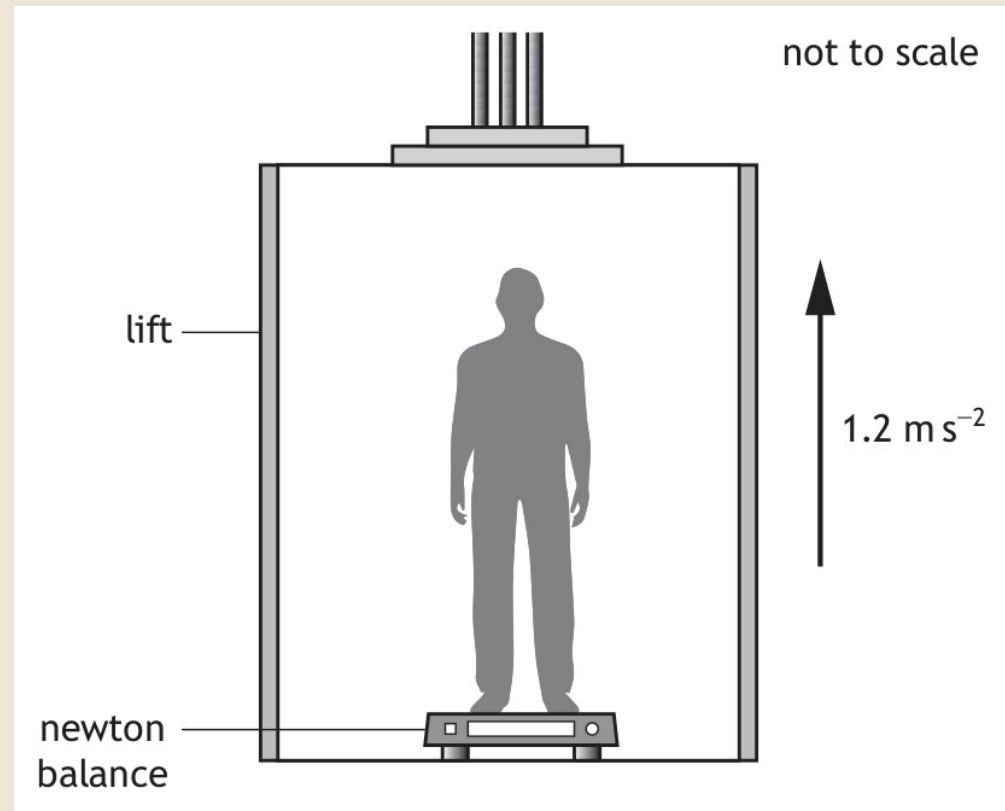
B 430 N

C 490 N

D 550 N

E 590 N

Answer D: $N - 490 = 50.0 \times 1.2$ — so $N = 550\text{ N}$



Spot the mistake. A 60 kg pupil is in a lift accelerating upwards at 2.0 m s^{-2} . What is wrong with this answer?

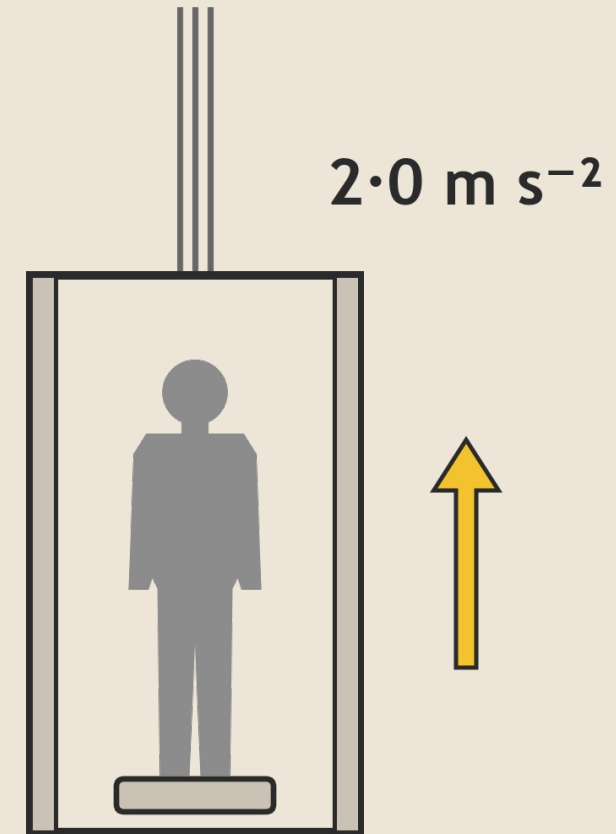


$$N = m(g - a)$$

$$N = 60 \times (9.8 - 2.0)$$

$$N = 468 \text{ N}$$

**a is upwards — so $N > W$: $N = 60 \times 11.8$
 $= 708 \text{ N}$**



Practical: Measure a Lift's Acceleration

You need

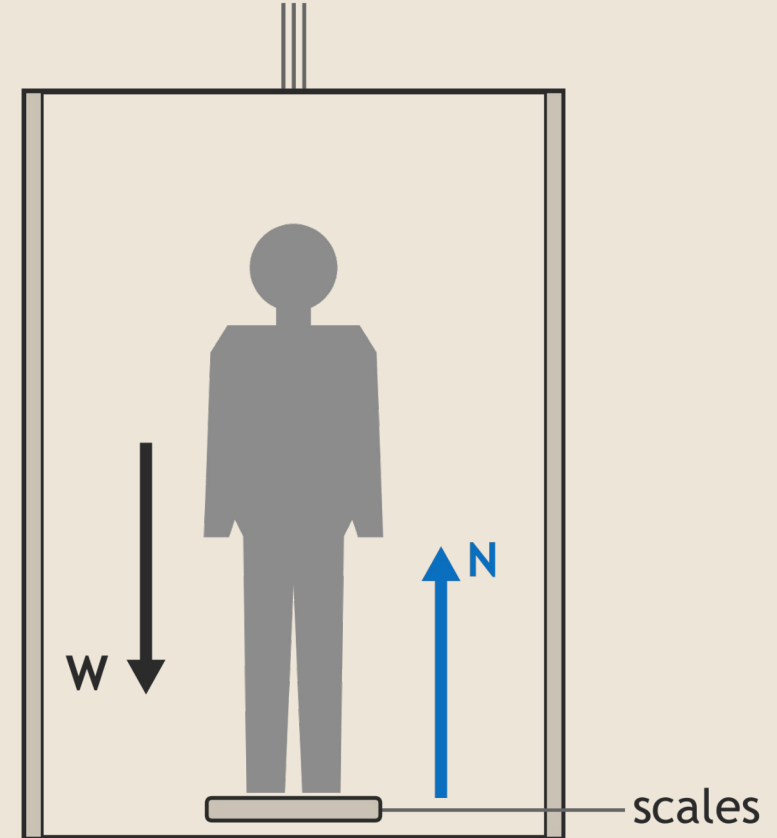
the lift, newton scales, a phone

Measure N

at rest, then on rides up and down

Film it

the reading changes in 1-2 s



Lift Practical



Lift practical sheet

Groups of 3: one rides, one films, one records.
Three rides up, three rides down.
Then check a with SPARKvue on a phone.



SAFETY An adult rides too. Stand still, scales flat.

From Readings to Acceleration

At rest

$$N = W, \text{ so } m = W \div 9.8$$

Each stage

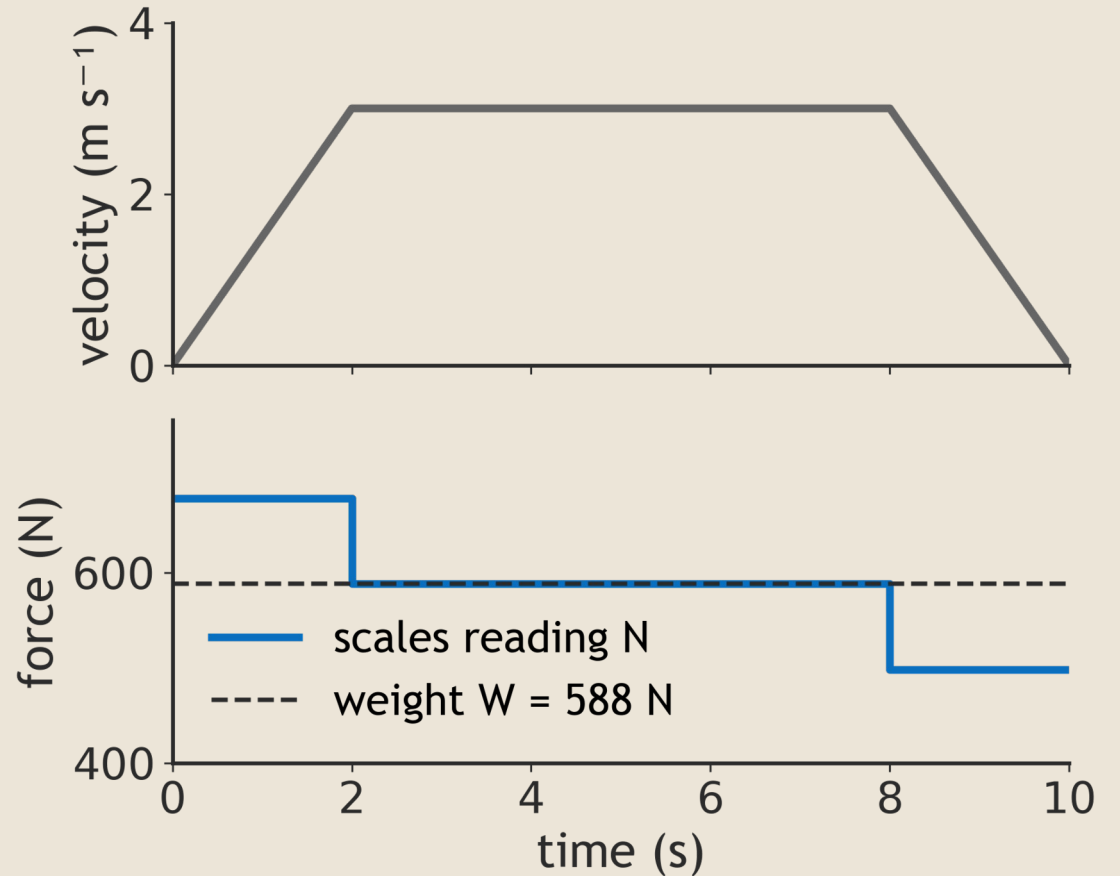
$$a = (N - W) \div m$$

Check

against SPARKvue

Evaluate

repeat; the scales lag



Plenary: Lifts



- 1 What do the scales read?
- 2 A lift slows down going up. Is N bigger or smaller than W ?
- 3 A 50 kg pupil. $a = 1.0 \text{ m s}^{-2}$ upwards. Calculate the reading.
- 4 Why film the scales display?

N , the upward push on you

Smaller – so a is downwards

540 N

It changes in 1-2 s – so too fast to read live

Look back — by the end of this part...



- 1 I can draw the forces on a person standing on scales in a lift
- 2 I can use $F = ma$ to find a scales reading or a lift's acceleration
- 3 I can describe how to measure a lift's acceleration with newton scales

2010 Higher Physics: scan the code and answer these.



Q2

Q16

Section A · motion, semiconductors

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 4

Friction and terminal velocity

A skydiver falls for a minute, but stops speeding up after about 15 s.

By the end of this part...

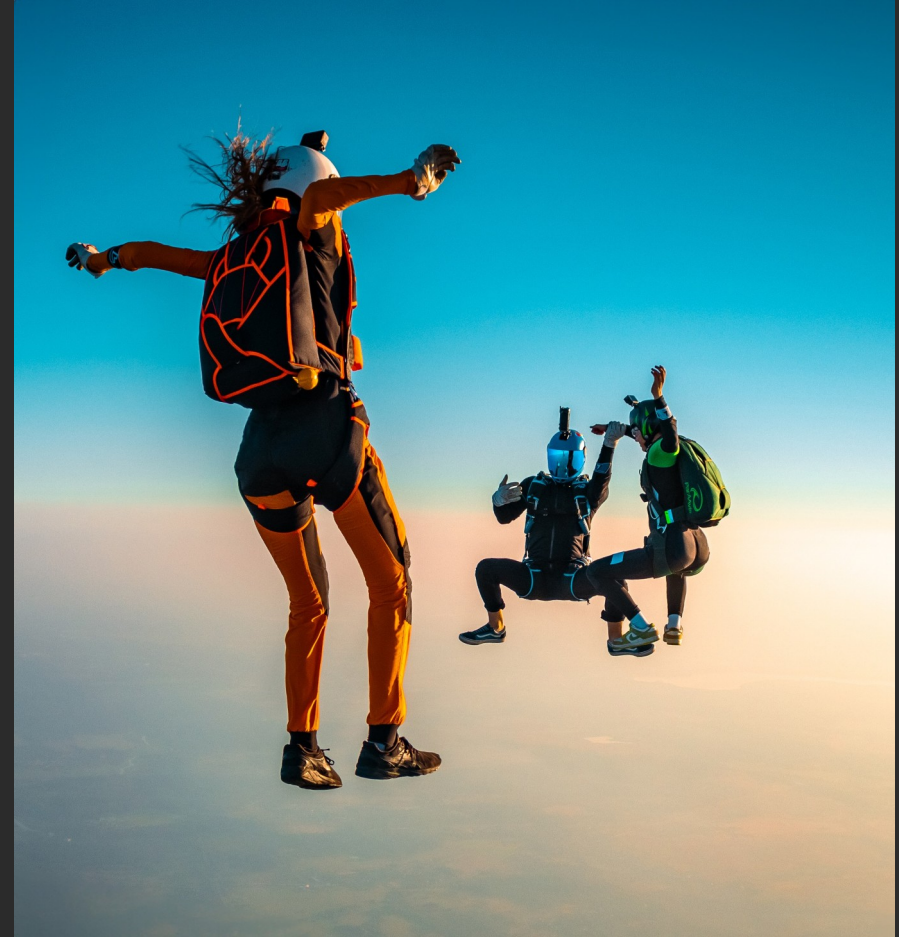


- 1 I can describe the effect of friction on a moving object
- 2 I can explain terminal velocity in terms of forces

Why does a skydiver stop speeding up?



Skydivers in free fall



A formation

Recap from N5: Friction

Friction

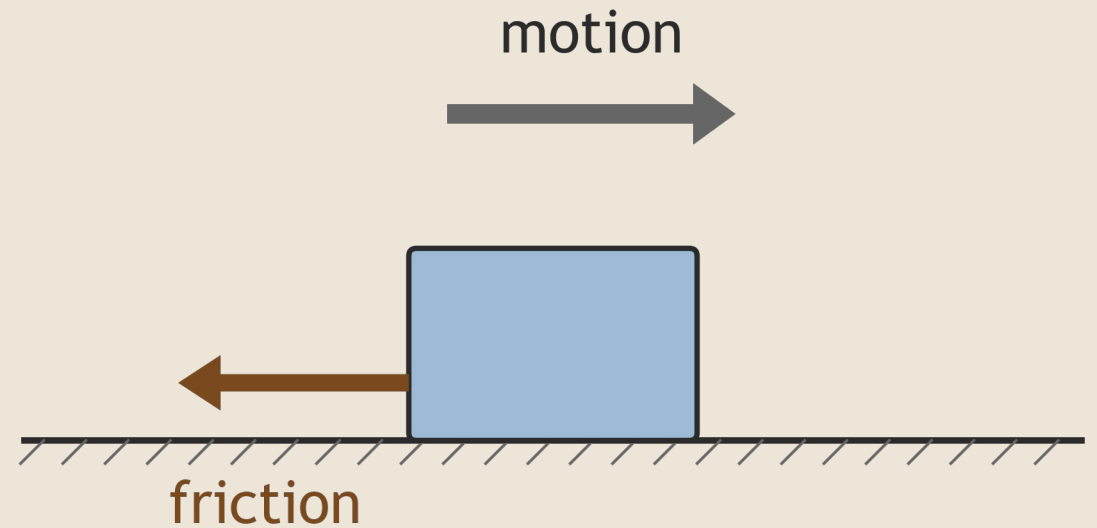
opposes motion between surfaces

Direction

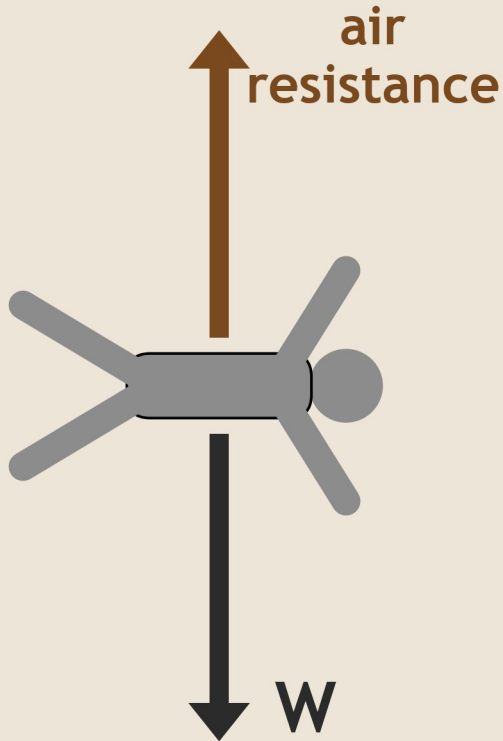
opposite to the motion

Air resistance

friction from the air; grows with speed



Terminal Velocity



$$v = 55 \text{ m s}^{-1}$$

1 Just jumped: only W .
 $a = 9.8 \text{ m s}^{-2}$.

2 Speeding up: air resistance grows.

3 Air resistance nearer W : a smaller.

4 Air resistance = W :
terminal velocity.

Explaining Terminal Velocity



Speed rises

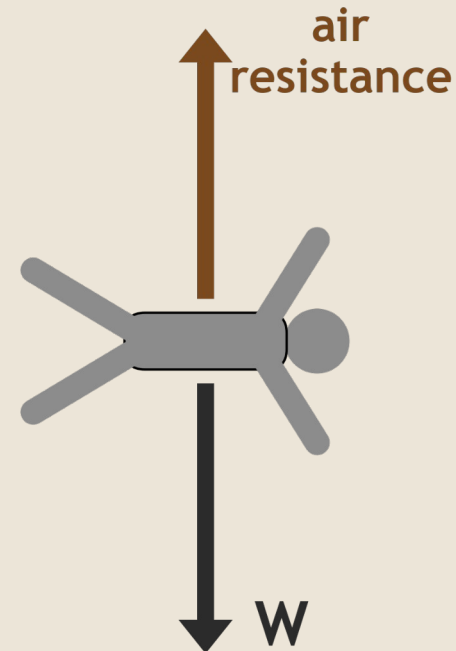
so air resistance rises

Until

air resistance = weight

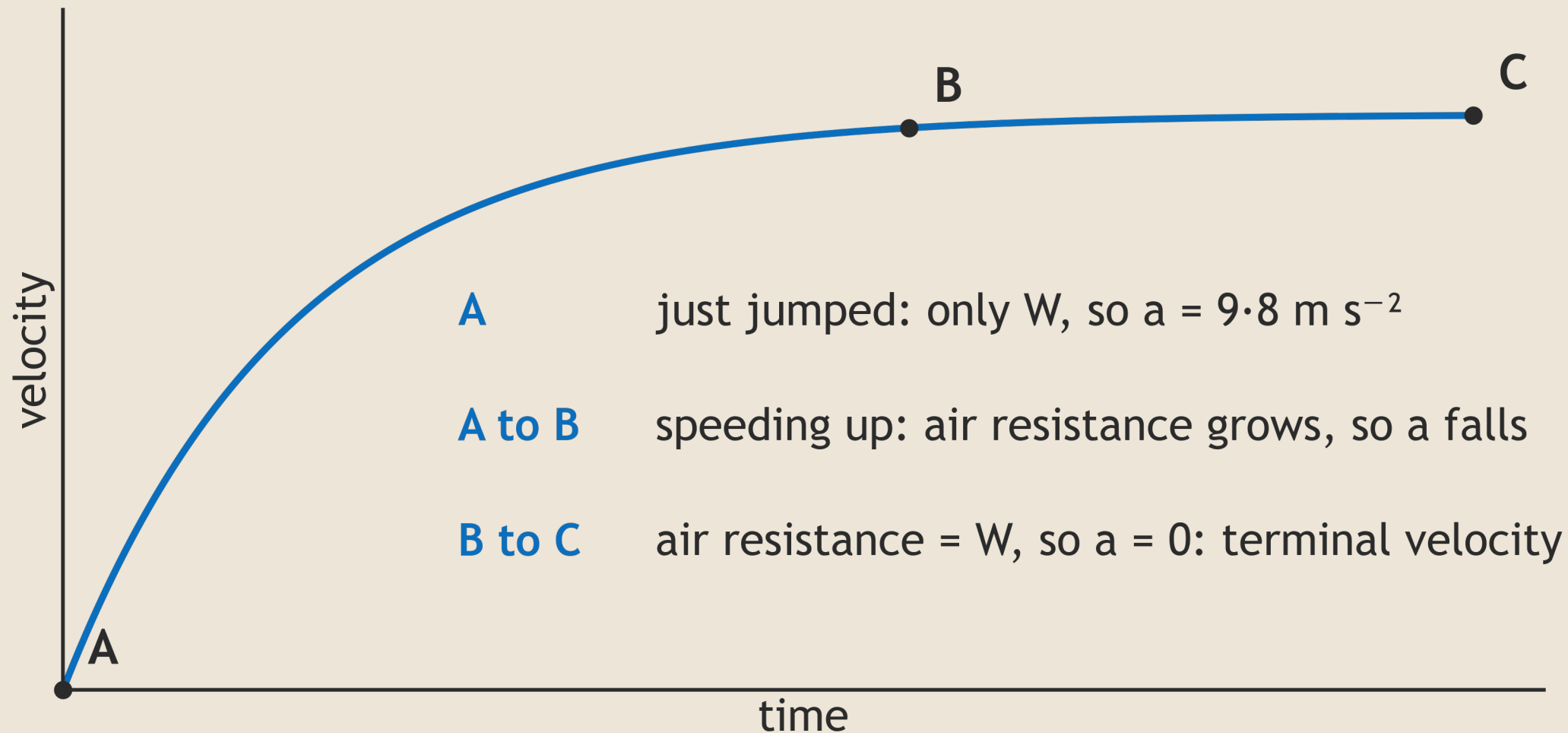
So

forces balanced: $a = 0$,
constant velocity



$$v = 55 \text{ m s}^{-1}$$

Terminal Velocity on a v-t Graph



Your Turn: Friction and Terminal Velocity



1

A 70 kg skydiver feels 500 N of air resistance. Calculate their acceleration.

2.7 m s^{-2}

2

A skydiver's weight is 750 N. At terminal velocity, state the air resistance.

750 N – so the forces are balanced

3

Explain why a skydiver's acceleration falls as they speed up.

Air resistance rises – so F_R falls

2026 Paper 1 Q2. A falling object travels at its terminal velocity because



- A the air resistance is less than the weight of the object
- B the air resistance is balanced by the weight of the object
- C the air resistance is greater than the weight of the object
- D the acceleration of the object is 9.8 m s^{-2}
- E the acceleration of the object is greater than 9.8 m s^{-2} .

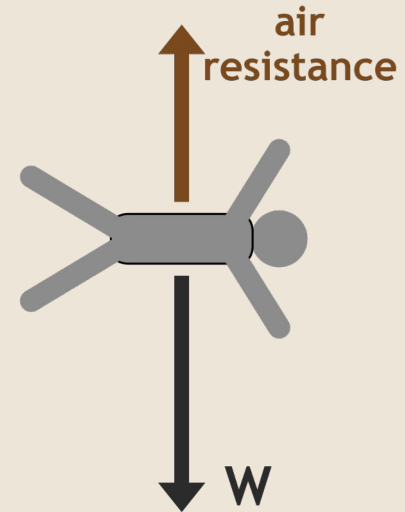
Answer B: balanced forces — so $a = 0$ and v is constant

Spot the mistake. A pupil writes this about a skydiver who has just jumped. What is wrong?



As the skydiver speeds up,
air resistance increases,
so the acceleration increases.

Air resistance opposes W – so the
unbalanced force and a decrease



$$v = 40 \text{ m s}^{-1}$$

Plenary: Friction and Terminal Velocity



1

A skydiver's air resistance is half their weight. State a .

4.9 m s^{-2}

2

At terminal velocity, what is the acceleration?

Zero

3

Why does a skydiver stop speeding up?

Air resistance = W — so
balanced

Look back — by the end of this part...



- 1 I can describe the effect of friction on a moving object
- 2 I can explain terminal velocity in terms of forces

2013 Revised Higher Physics: scan the code and answer these.



Q2

Q4

Q20

Section A · motion, forces, p-n

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 5

Falling with air resistance

One graph tells the whole story of a skydive.

By the end of this part...



- 1 I can interpret a velocity-time graph for a falling object
- 2 I can link each part of the graph to the forces

A Skydiver's v-t Graph



A to B

air resistance grows

B to C

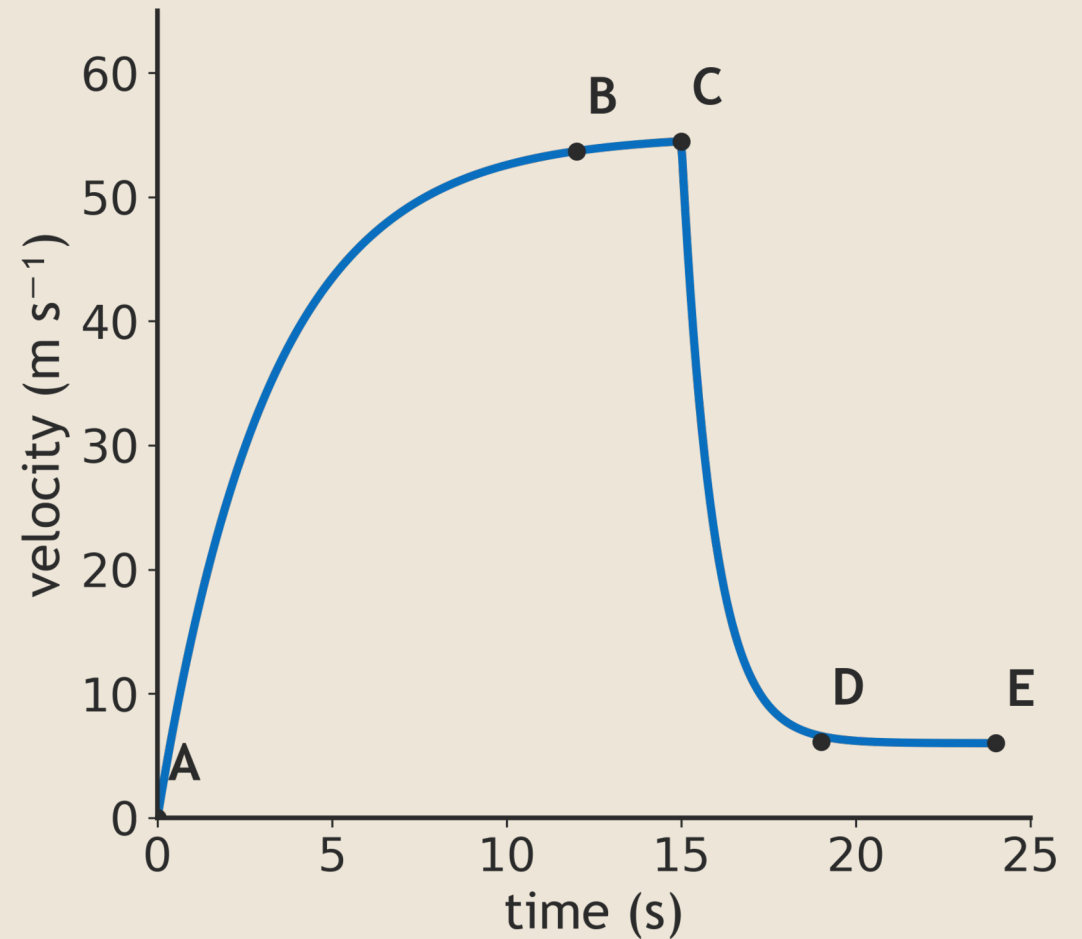
terminal velocity

C to D

air resistance $>$ W : slows

D to E

lower terminal velocity



The Gradient Is the Acceleration

Steep

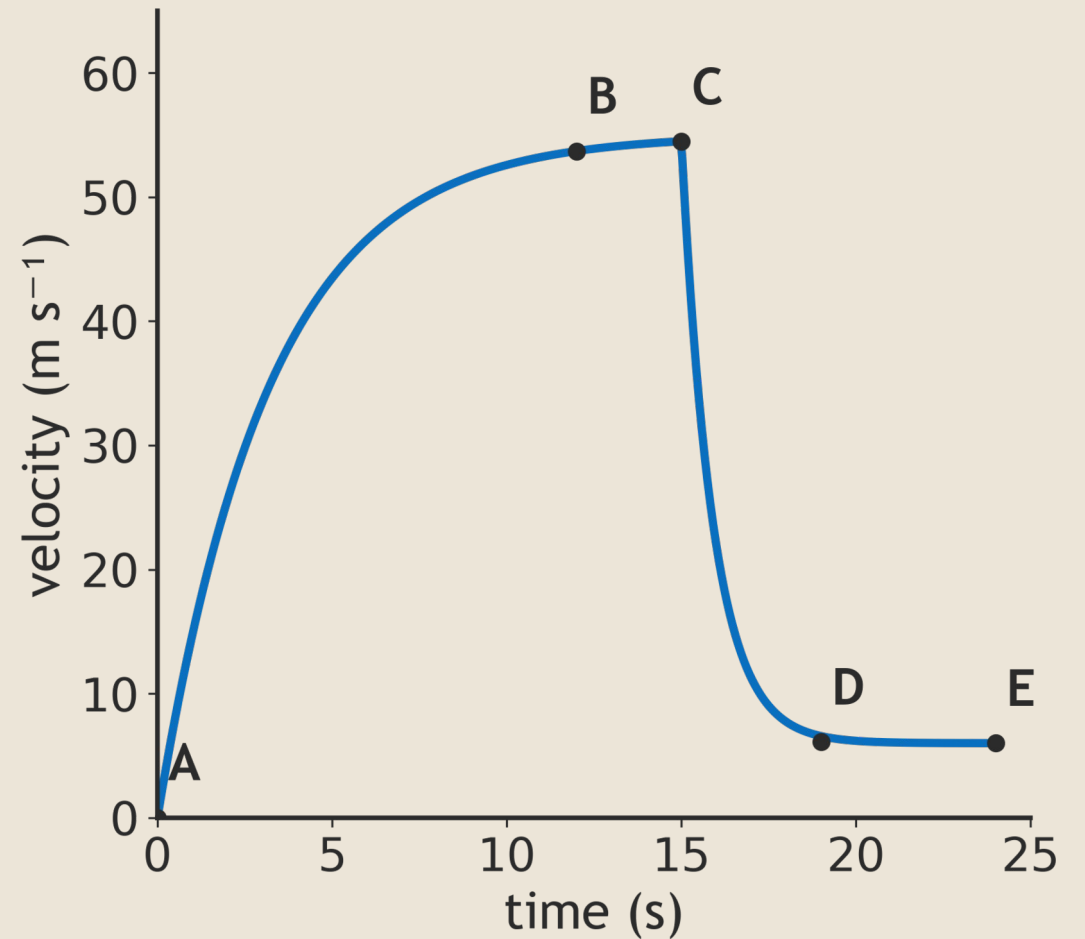
large a : W much bigger than air resistance

Gradient falls

a falls as air resistance grows

Flat

$a = 0$: forces balanced



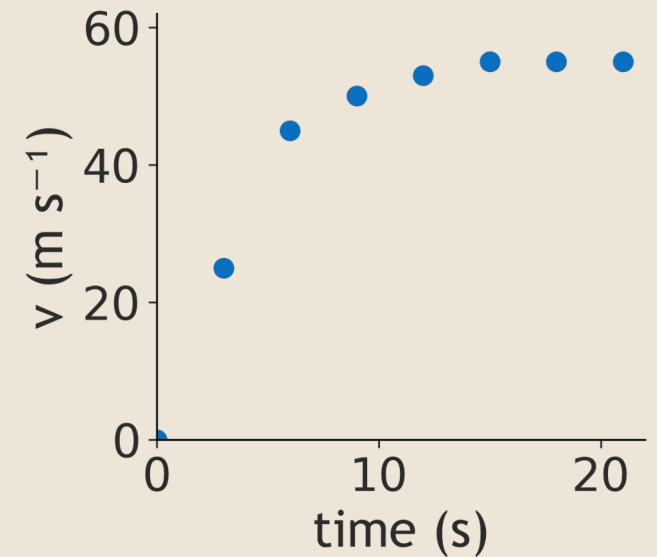
Worked Example: A Skydiver's Data

A skydiver's speed is recorded every 3 s (graph). Explain the forces from 15 s to 21 s.

2

v is constant (55 m s^{-1}), so $a = 0$ (1)

air resistance = weight, so balanced (1)



Your Turn: Skydiver Graphs



1

Sketch the v - t graph of an object falling from rest to terminal velocity.

A curve that levels off

2

State what the gradient of a v - t graph shows.

The acceleration

3

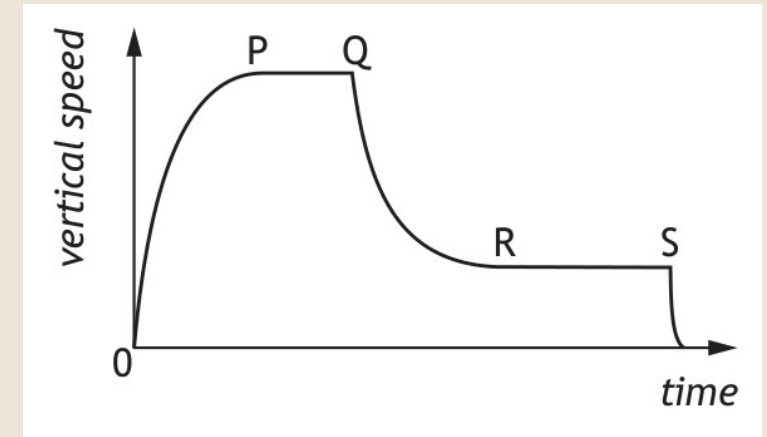
A parachute opens. Compare air resistance with W .

Bigger – so the skydiver slows

2017 Section 1 Q3. A skydiver's vertical speed against time. Which statements are correct?



- I a is greatest from P to Q.
- II Q to R: air resistance $<$ weight.
- III R to S: the forces are balanced.



A I only B II only **C III only** D I, II only E I, II, III

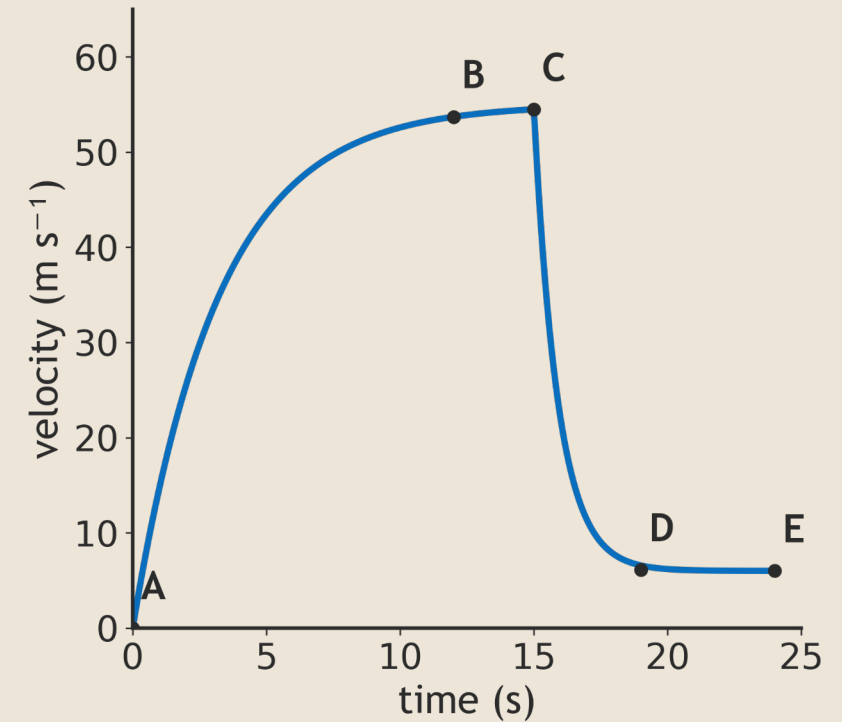
Answer C: P to Q is flat ($a = 0$) and Q to R slows (air resistance $>$ W)
– so only III

Spot the mistake. A pupil matches the a-t graph to a skydiver's v-t graph. What is wrong?



The v-t graph rises, then levels off,
so the a-t graph also rises,
then levels off.

a is the gradient — so a starts at 9.8 m s^{-2} and falls to 0



Plenary: Falling with Air Resistance



1

On a v-t graph, what does a flat line mean?

Terminal velocity — $a = 0$

2

When the parachute opens, which way is the unbalanced force?

Up

3

Is the second terminal velocity higher or lower?

Lower

Look back — by the end of this part...



- 1 I can interpret a velocity-time graph for a falling object
- 2 I can link each part of the graph to the forces

2014 Revised Higher Physics: scan the code and answer these.



Q2

Q17

Q20

Section A · motion, p.d., uncertainty

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 6

Resolving vectors

Split one force into two that are easier to use.

By the end of this part...



- 1 I can resolve a vector into two perpendicular components

Resolving a Force



Horizontal

$$F_h = F \cos \theta$$

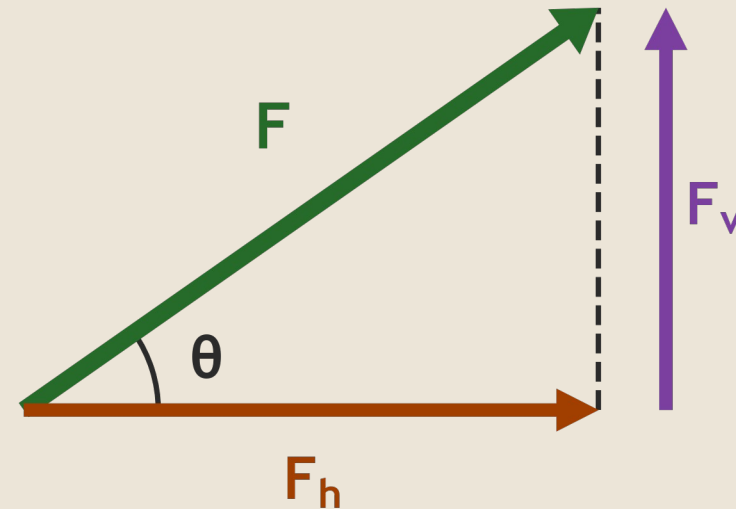
Vertical

$$F_v = F \sin \theta$$

θ

measured from the
horizontal

F = the force being resolved



Resolving: The Method



1 First find F_h , using:

$$F_h = F \cos \theta$$

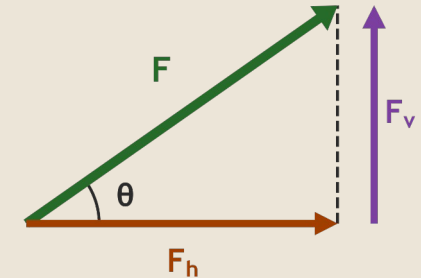
2 Then find a , using:

$$F_h - \text{friction} = ma$$

F = the pull, at θ above the horizontal

F_h = the part of F along the floor

No friction: $F_h = ma$



Worked Example: Pulling at an Angle

A 15 N force at 60° above the horizontal pulls a 2.5 kg box along a smooth floor. Calculate its acceleration.

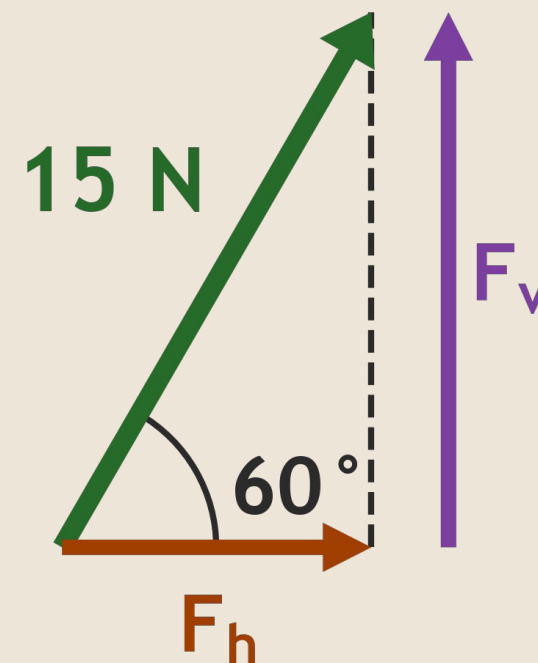
3

$$F = ma \quad (1)$$

$$(F_h = 15 \cos 60^\circ = 7.5 \text{ N})$$

$$7.5 = 2.5 \times a \quad (1)$$

$$a = 3.0 \text{ m s}^{-2} \quad (1)$$



Worked Example: A Sledge with Friction

A rope pulls a 15 kg sledge with 80 N at 35° to the horizontal. Friction is 20 N. Calculate its acceleration.

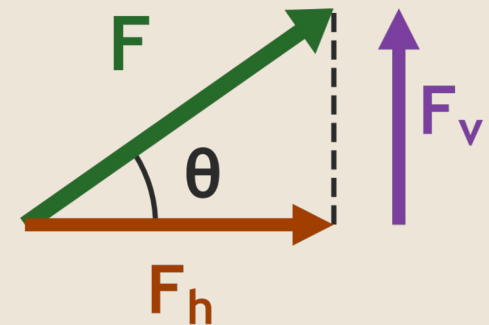
3

$$F = ma \quad (1)$$

$$(F_h = 80 \cos 35^\circ = 65.5 \text{ N})$$

$$65.5 - 20 = 15 \times a \quad (1)$$

$$a = 3.0 \text{ m s}^{-2} \quad (1)$$



Your Turn: Resolving



1

A 20 N force at 30° above the horizontal pulls a 4.0 kg block on a smooth floor. Calculate its acceleration.

$$4.3 \text{ m s}^{-2}$$

2

A 12 N force acts at 25° above the horizontal. Calculate both components.

$$F_h = 11 \text{ N}, F_v = 5.1 \text{ N}$$

3

θ is from the horizontal. Which function gives the vertical part?

$$\sin \theta$$

2024 Paper 1 Q3. A box hangs from a rope. A horizontal force F holds it stationary as shown. The weight of the box is 4.9 N. The tension T in the rope is



A 1.7 N

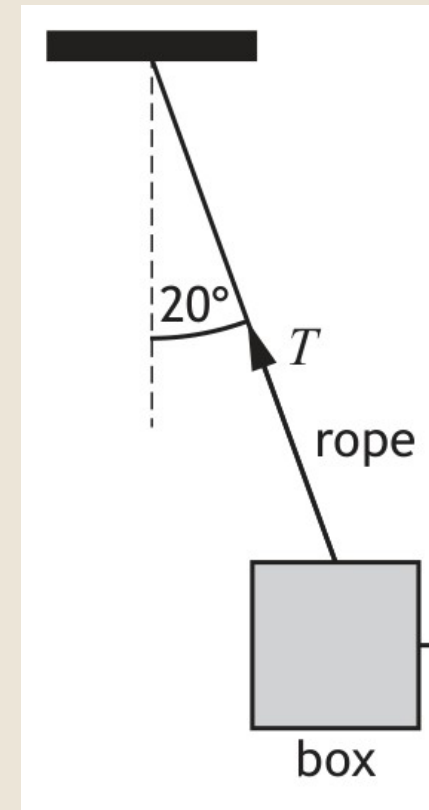
B 4.6 N

C 4.9 N

D 5.2 N

E 14 N

Answer D: $T \cos 20^\circ = 4.9 \text{ N}$
— so $T = 5.2 \text{ N}$



Spot the mistake. A 6.0 N weight hangs from a rope at 30° to the vertical, held by a horizontal force. What is wrong?

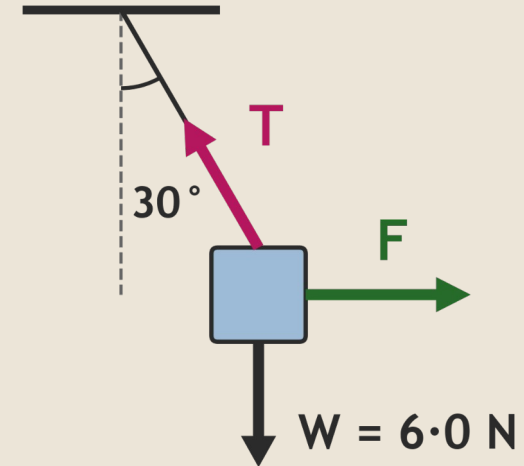


$$T = W \cos 30^\circ$$

$$T = 6.0 \times \cos 30^\circ$$

$$T = 5.2 \text{ N}$$

T's vertical part balances W: $T \cos 30^\circ$
 $= 6.0$ — so $T = 6.9 \text{ N}$



Plenary: Resolving



1

Resolve 10 N at 45° above the horizontal.

7.1 N and 7.1 N

2

Which component uses $\cos, \theta$ from the horizontal?

F_h (horizontal)

3

Name the topic in ODU 1 that resolved a velocity.

Projectiles

Look back — by the end of this part...



- 1 I can resolve a vector into two perpendicular components

2009 Higher Physics: scan the code and answer these.



Q2

Q9

Q18

Section A · motion, electricity

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 7

Forces on a slope

On a slope, only part of the weight pulls you down.

By the end of this part...



- 1 I can resolve weight on a slope, parallel and normal to the surface
- 2 I can solve problems on a slope, including friction

Weight on a Slope



Down the slope

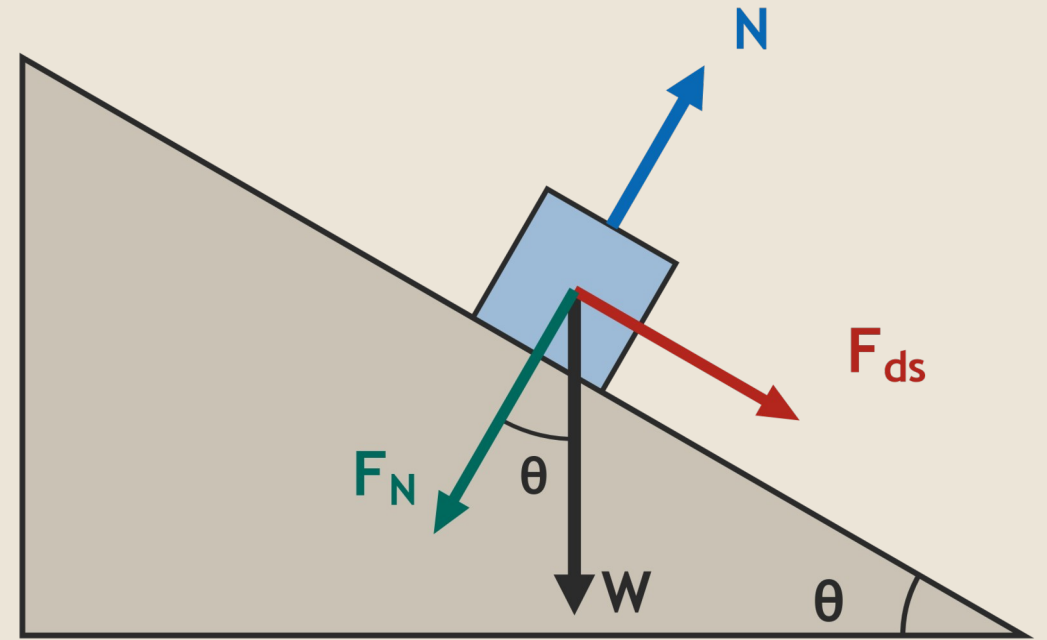
$$F_{ds} = mg \sin \theta$$

Into the slope

$$F_N = mg \cos \theta, \text{ balanced by } N$$

θ

the slope angle, and between W and the normal



Slopes: The Method



1 First find F_{ds} , using:

$$F_{ds} = mg \sin \theta$$

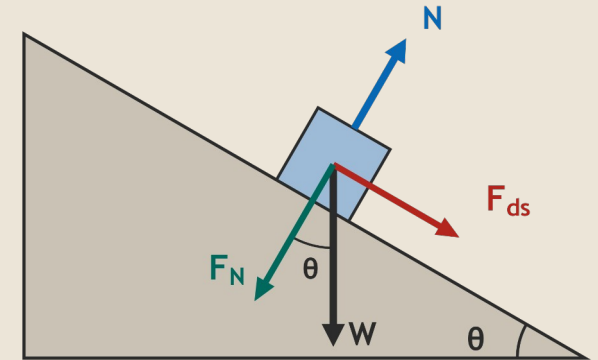
2 Then find a , using:

$$F = ma$$

F = unbalanced force, e.g. F_{ds} – friction

F_{ds} = part of the weight down the slope

Constant speed: $a = 0$, forces balanced



Worked Example: Pulled Up a Slope

A 50 kg crate is pulled up a 20° slope at constant speed.
Friction is 60 N. Calculate the pulling force.

4

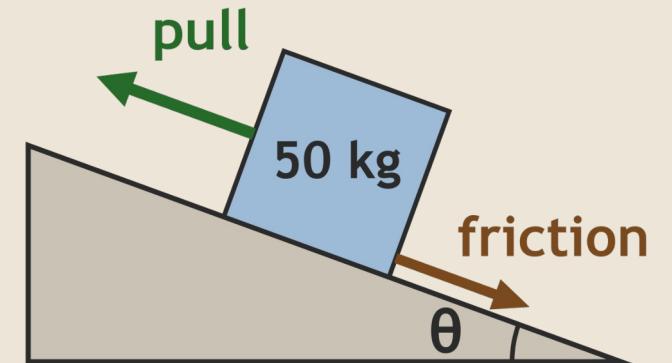
$$F_{ds} = mg \sin \theta \quad (1)$$

$$F_{ds} = 50 \times 9.8 \times \sin 20^\circ \quad (1)$$

$$F_{ds} = 168 \text{ N} \quad (1)$$

(constant speed, so forces balanced)

$$\text{pull} = 168 + 60 = 230 \text{ N} \quad (1)$$



Worked Example: Sliding Down

A 4.0 kg block slides down a 30° slope. Friction is 5.0 N.
Calculate its acceleration.

5

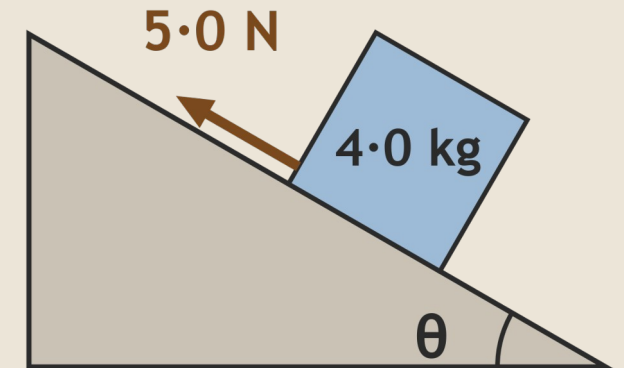
$$F_{\text{ds}} = mg \sin \theta \quad (1)$$

$$F_{\text{ds}} = 4.0 \times 9.8 \times \sin 30^\circ = 19.6 \text{ N} \quad (1)$$

$$F = ma \quad (1)$$

$$19.6 - 5.0 = 4.0 \times a \quad (1)$$

$$a = 3.7 \text{ m s}^{-2} \quad (1)$$



Watch Out: Friction on a Slope

Moving up

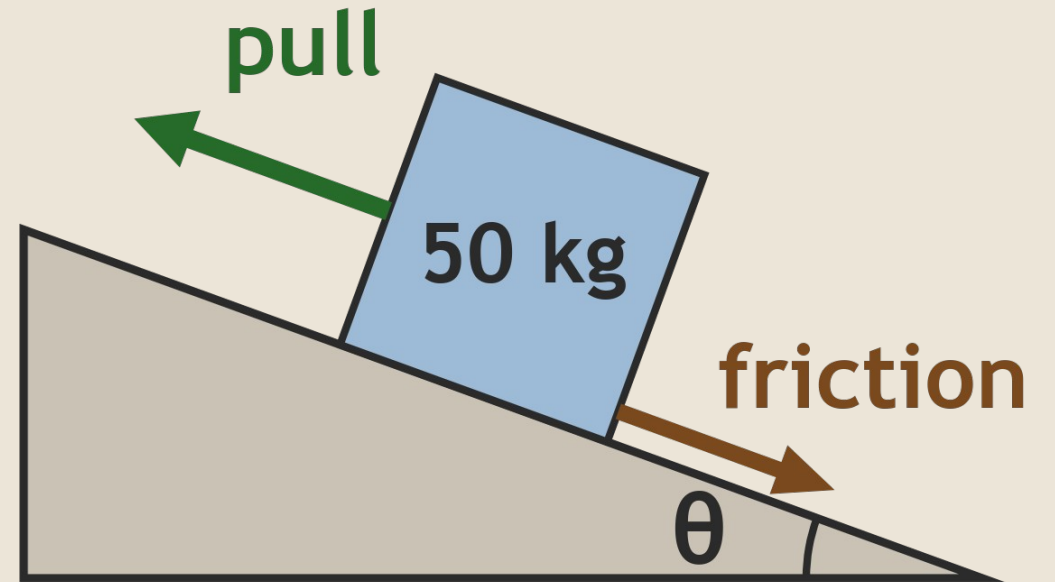
friction acts down: add it

Moving down

friction acts up: subtract it

Always

friction opposes the motion



Your Turn: Slopes



1

A 3.0 kg box slides down a smooth 25° slope. Calculate its acceleration.

$$4.1 \text{ m s}^{-2}$$

2

State which component of the weight N balances.

$$F_N = mg \cos \theta$$

3

A trolley is pushed up a slope at constant speed. $F_{ds} = 50 \text{ N}$, friction 20 N. Calculate the push.

70 N – so add the friction

2026 Paper 1 Q3. A cyclist and bicycle (70.0 kg) freewheel down a 15° slope. The component of their combined weight down the slope is



A 18 N

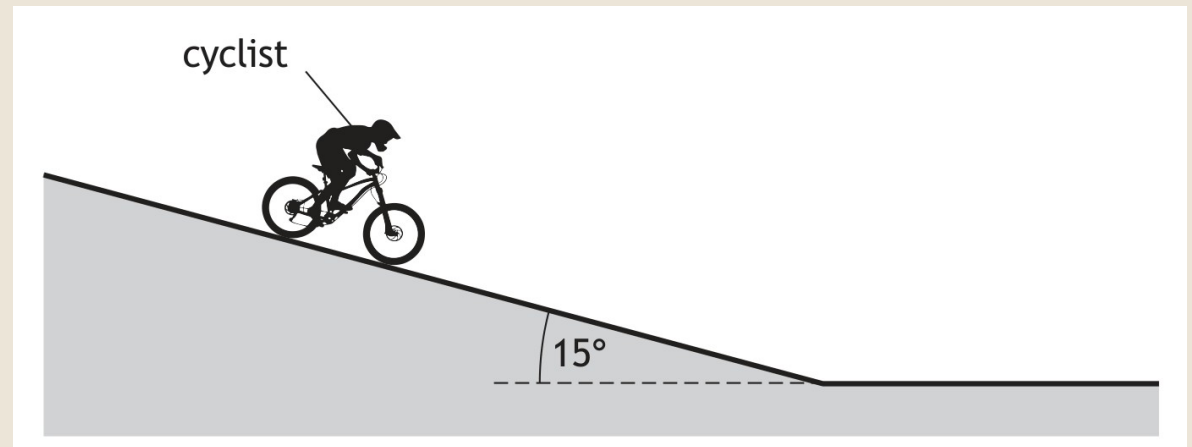
B 68 N

C 180 N

D 660 N

E 690 N

Answer C: $70.0 \times 9.8 \times \sin 15^\circ$ — so 180 N



2022 Paper 1 Q3. A 3.0 kg package moves down a ramp on the Moon ($g = 1.6 \text{ N kg}^{-1}$). Its weight's component parallel to the ramp is:



A 0.89 N

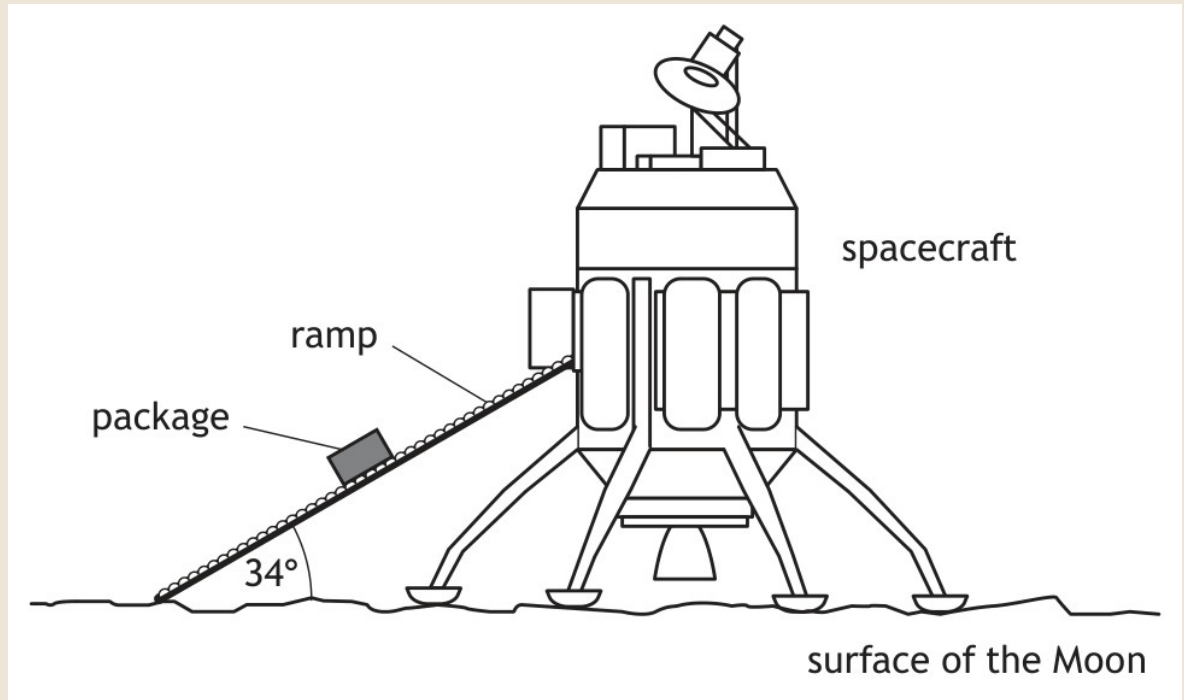
B 2.7 N

C 4.0 N

D 4.8 N

E 16 N

Answer B: $3.0 \times 1.6 \times \sin 34^\circ$ — so 2.7 N



Spot the mistake. A 5.0 kg box is pushed up a smooth 30° slope at constant speed. What is wrong?

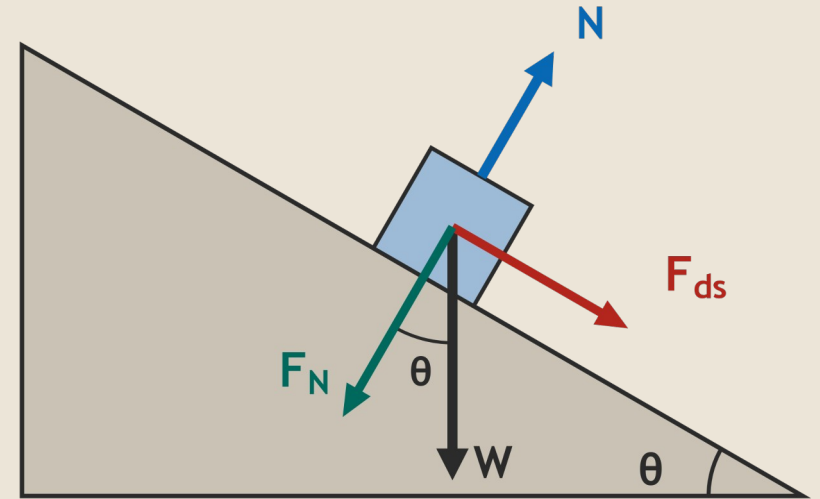


$$F = W$$

$$F = 5.0 \times 9.8$$

$$F = 49 \text{ N}$$

Only F_{ds} acts down the slope — so $F = 49 \times \sin 30^\circ = 25 \text{ N}$



Plenary: Forces on a Slope



1

Which component of weight pulls a box down a slope?

$$F_{ds} = mg \sin \theta$$

2

A box slides down. Which way does friction act?

Up the slope

3

Pushed up at constant speed: push = ?

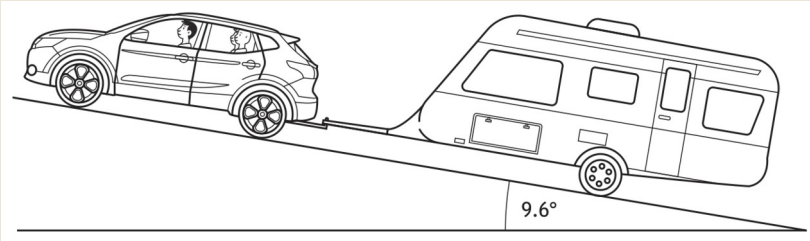
$$F_{ds} + \text{friction}$$

Look back — by the end of this part...

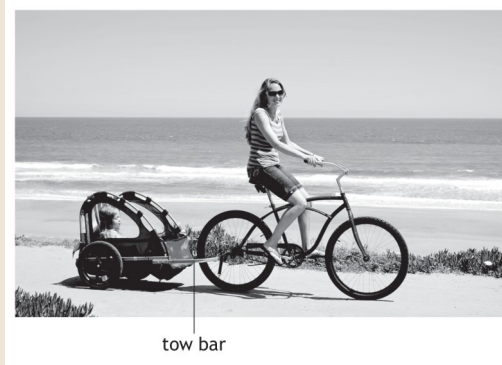


- 1 I can resolve weight on a slope, parallel and normal to the surface
- 2 I can solve problems on a slope, including friction

Forces in the Exam



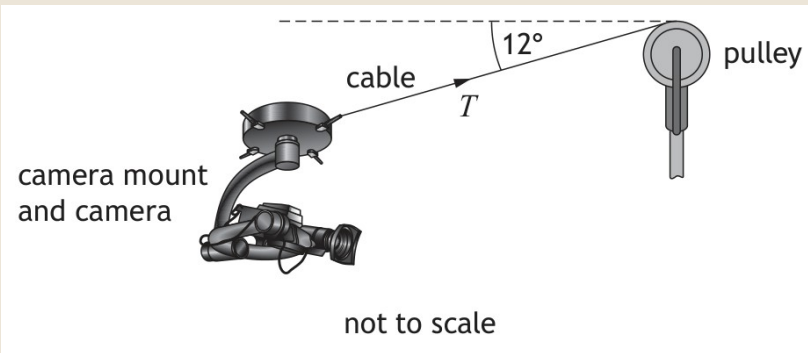
2024 Paper 2 Q2



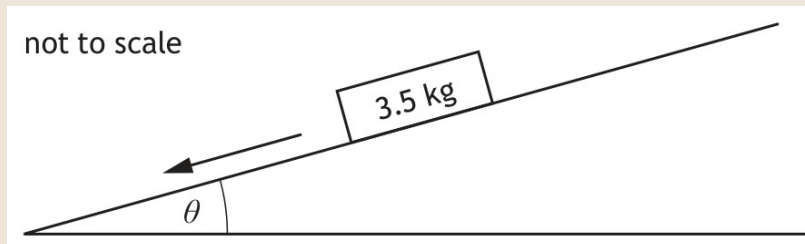
2023 Paper 2 Q2



Past papers



2025 Paper 2 Q1(b)



2025 Paper 1 Q4

Kept for your
test: 2018 P1
Q4, 2016 Q3,
2017 Q2.

Brain Dump



In your jotter

Write everything you remember about forces.
On your own, 2 minutes. No notes.
Then check it against the plenaries.

Extension

EXTRA PRACTICE



1 A car's engine force is constant.
Explain why its acceleration falls as it speeds up.

Air resistance rises –
**so the unbalanced
force falls**

2 Explain why the normal force on a slope is less than the weight.

N balances F_N – so $N < W$

LOOKING AHEAD

Next: Energy and Power

Every force that moves something does work.

Course specification — 1 of 3

- Use of vector addition and appropriate relationships to solve problems involving balanced and unbalanced forces, mass, acceleration, and gravitational field strength.
- Knowledge of the effects of friction on a moving object (no reference to static and dynamic friction).
- Explanation, in terms of forces, of an object moving with terminal velocity.

Course specification – 2 of 3

- Interpretation of velocity-time graphs for a falling object when air resistance is taken into account.
- Use of Newton's first and second laws to explain the motion of an object.
- Use of free body diagrams and appropriate relationships to solve problems involving friction and tension.

Course specification — 3 of 3

- Resolution of a vector into two perpendicular components.
- Resolution of the weight of an object on a slope into component forces parallel and normal to the surface of the slope.

Energy and power, the last bullet, is deck 2b.

Credits (1 of 3)

Starter: 2012 Revised Higher Physics, © Qualifications Scotland (SQA); 2015 Higher Physics, © Qualifications Scotland (SQA); 2014 Higher Physics, © Qualifications Scotland (SQA); 2010 Higher Physics, © Qualifications Scotland (SQA); 2013 Revised Higher Physics, © Qualifications Scotland (SQA); 2014 Revised Higher Physics, © Qualifications Scotland (SQA); 2009 Higher Physics, © Qualifications Scotland (SQA)

2025 Paper 1 Q3: © Qualifications Scotland (SQA). Higher Physics 2025, Paper 1, Question 3.

2015 Section 1 Q5: © Qualifications Scotland (SQA). Higher Physics 2015, Section 1, Question 5.

The big question (A lift): © David Anstiss, CC BY-SA 2.0, via Geograph

The big question (A drop tower ride): Photo: Stefan Scheer, CC BY-SA 4.0, via Wikimedia Commons

Credits (2 of 3)

2015 Section 1 Q4: © Qualifications Scotland (SQA). Higher Physics 2015, Section 1, Question 4.

2022 Paper 1 Q5: © Qualifications Scotland (SQA). Higher Physics 2022, Paper 1, Question 5 (figure and text).

The big question (Skydivers in free fall):
Photo: Rstpch, CC BY-SA 4.0, via Wikimedia Commons

The big question (A formation): Photo:
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2026 Paper 1 Q2: © Qualifications Scotland (SQA). Higher Physics 2026, Paper 1, Question 2.

2017 Section 1 Q3: © Qualifications Scotland (SQA). Higher Physics 2017, Section 1, Question 3.

2024 Paper 1 Q3: © Qualifications Scotland (SQA). Higher Physics 2024, Paper 1, Question 3.

2026 Paper 1 Q3: © Qualifications Scotland (SQA). Higher Physics 2026, Paper 1, Question 3.

Credits (3 of 3)

2022 Paper 1 Q3: © Qualifications Scotland (SQA). Higher Physics 2022, Paper 1, Question 3.

Forces in the Exam: © Qualifications Scotland (SQA). Higher Physics 2023 Paper 2 Q2, 2024 Paper 2 Q2, 2025 Paper 1 Q4 and Paper 2 Q1.

Course specification — 1 of 3: © Qualifications Scotland (SQA). Higher Physics Course Specification v3.0, p.7

Course specification — 2 of 3: © Qualifications Scotland (SQA). Higher Physics Course Specification v3.0, p.7

Course specification — 3 of 3: © Qualifications Scotland (SQA). Higher Physics Course Specification v3.0, p.7